

CHE 321 – Chemical Reaction Engineering

Fall 2019

Homework Assignment – 1 (50 points)

Due: 12 PM, 23rd September, 2019

Problem-1

Consider the following elementary reaction,



The rate equation for the above reaction is given as,

$$r = k C_A C_B \quad 1$$

where r denotes the rate of the reaction ($\text{mol/m}^3\text{s}$), k denotes the rate constant ($(\text{mol/m}^3)^{-1}\text{s}^{-1}$) and C_A , C_B denotes the concentrations of A and B (mol/m^3) respectively. The value of k can be determined from the Arrhenius equation which is,

$$k = A \exp\left(-\frac{E}{RT}\right) \quad 2$$

where A is the frequency factor ($(\text{mol/m}^3)^{-1}\text{s}^{-1}$), E is the activation energy (J/mol K), R is the gas constant (J/mol) and T is the temperature (K). The values of A and E can be determined by doing a linear regression of the experimental data.

It is possible to theoretically estimate the value of k from several theories such as collision theory, transition state theory, UBI-QEP theory etc. In this assignment, you shall **derive the expression for the rate constant from the collision theory.**

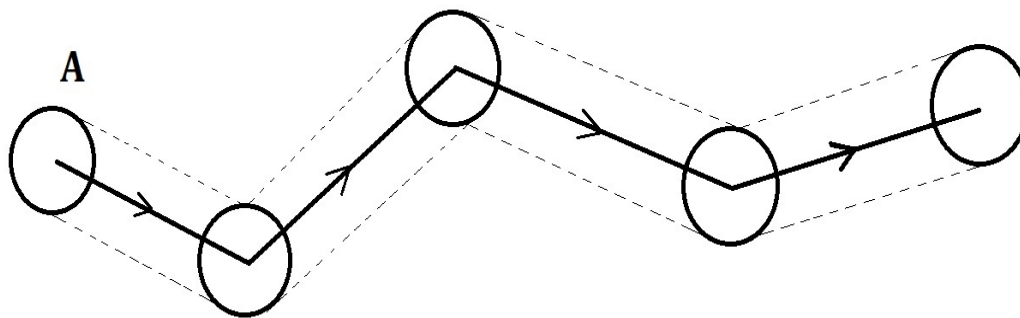
Let n_A and n_B denote the number of molecules of A and B per unit volume which are known as the number density of A and B. The rate of disappearance of A is given as

$$-\frac{dn_A}{dt} = Z_{AB} P \quad 3$$

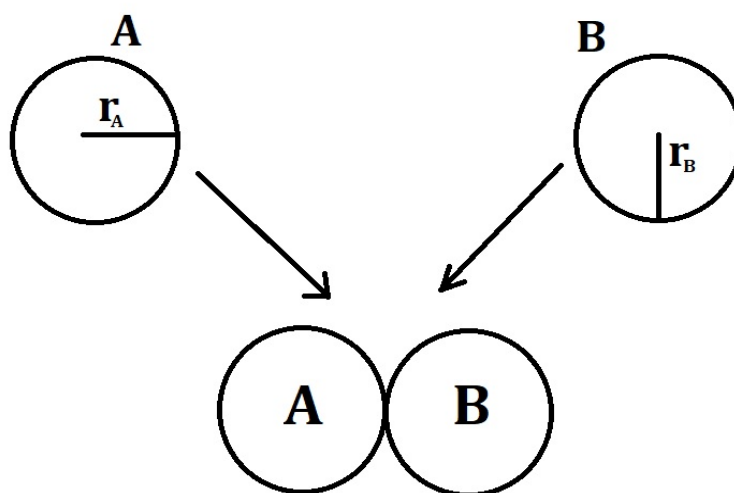
where Z_{AB} is the collision frequency (m^3s^{-1}) (i.e.,) the number of collisions per unit volume per unit time and P denotes the probability that the collision will provide enough energy to overcome the activation barrier.

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Consider the molecule A traveling in a random path. While traveling in this path the molecule A sweeps an imaginary area denoted by the dotted lines in the above figure. Similarly, the motion of molecule B can be denoted by an imaginary area.



For an effective collision, the molecules of A and B must come closer and an imaginary moving tube can be constructed indicating the area in which the collision is effective. If a molecule of A meets a molecule of B in the swept area, then a collision occurs and outside the swept area there will not be a collision. This imaginary moving tube can be constructed for each pair of A and B molecules. In the above figure, r_A and r_B denote the radii of A and B respectively.

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The motion of the molecules of A and B can be approximated by the average velocity,

$$v_{AB} = \left(\frac{8k_B T}{\pi \mu} \right)^{\frac{1}{2}} \quad 4$$

where k_B denotes the Boltzmann constant and $\mu = \frac{m_A m_B}{m_A + m_B}$ is the reduced mass, m_A and m_B denote the mass of one molecule of A and B respectively. The above expression for the average velocity is obtained from the kinetic theory of gases.

- 1) Obtain the expression for the collision frequency Z_{AB} and explain the procedure. (4 points)**

Hint: The collision frequency can be found by the product of the no. of molecules involved and the traveling swept volume per unit time where the collision occurs.

The probability that the collision will provide enough energy to overcome the activation barrier can be found from the statistical mechanics and it is given by

$$P = \exp\left(-\frac{E^*}{k_B T}\right)$$

where E^* is the energy required to overcome the electron cloud repulsion between A and B (J). This energy must be overcome for the collision to occur.

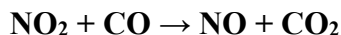
- 2) Obtain the rate equation from collision theory in terms of C_A and C_B as given by equation 1. (6 points)**

Hint: The rate constant from collision theory will resemble the Arrhenius equation.

Collision theory can be applied only for the simple molecules. For complicated systems, the steric effect plays a prominent role.

- 3) What modification can be done in the rate equation which you have derived to extend it to a complicated system? (4 points)**

Consider the following elementary reaction,



The radii of NO_2 and CO molecules are 165 and 188 pm respectively. The energy required to overcome the electron cloud repulsion between NO_2 and CO is 3×10^{-5} fJ. Assume that the steric effect does not play a major role in this reaction.

- 4) Predict the rate constant for the above reaction from collision theory at temperatures 300, 350, 400, 450 and 500 K and explain the effect of temperature on rate constant. (6 points)**

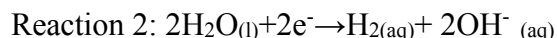
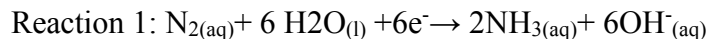
Problem-2:

- 1) Electrochemical Production of Ammonia (NH_3) and Hydrogen (H_2) in Alkaline Solutions (8 points)**

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In an electrochemical batch reactor with a total volume of 100 L, the following two irreversible reactions take place at 20°C in alkaline solution.



The pH is kept constant throughout the reaction and H_2O can be considered in excess, such that reaction invariant is not dependent on the concentration of H_2O and pH. The species of interest are N_2 , NH_3 , and H_2 dissolved in water. Initially, the reaction mixture contains only N_2 at the solubility limit of 0.0019 gm/(100gm water). Assume concentration of H_2O to be 55 moles/liter. Solubility limit of hydrogen is 0.00016 gr gas/(100 gr water).

- 1) *Determine the physical range of molar extents for reactions 1 and 2. What can be said about the rate of reaction 2 when the maximum molar extent is achieved? [Hint: sparingly soluble compounds can phase separate when their concentration increases beyond the saturation (or solubility) limit]*
- 2) *Find the reaction invariant for the species involved in reactions 1 and 2, and plot it using Matlab for the physical range of molar extents determined above.*
- 3) *How does the reaction invariant change when H_2 is also present initially at the solubility limit of 0.00016 gm gas/(100 gm water) along with N_2 ?*

2) Electrochemical Production of Ammonia (NH_3) and Hydrogen (H_2) in Acidic Solutions (8 points)

Now consider the previous reactions 1 and 2 in a semi-batch reactor where N_2 gas is constantly supplied to a fixed-volume batch of acidic solution, such that the solution is always saturated with N_2 throughout the course of reactions 1 & 2. Here the species of interest are H^+ , NH_3 , and H_2 which are dissolved in water. Initially, there are no products in the acidic solution whose pH is 1.

- 1) *Write the reactions 1 & 2 of acidic solution involving H^+ ions.*
- 2) *Find the reaction invariant for the species in the new reactions and plot it in Matlab for a physical range of molar extents.*

3) Economic Analysis of Ammonia and Hydrogen Production (4 points)

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Consider the cost of N_2 to be \$10/kg (\$70 for 40L, 135 bar cylinder), NH_3 to be \$8/kg, H_2 to be \$27/kg (\$500 for 50L, 300 bar cylinder) and HCl (acid) to be \$0.09 /kg (\$27 for 270 kg). Obtain the total material cost for the range of molar extents for reactions in alkaline and acidic solution and discuss the economic feasibility of the NH_3 production in alkaline vs an acidic solution.

Problem-3: Copper and sulfuric acid react to give sulfur dioxide and copper sulfate.

- 1) Write and balance the chemical reaction. **(2 points)**
- 2) How much copper and how much 94% acid are needed to make 1 kg of sulfur dioxide? **(4 points)**
- 3) If 1 kg of copper dissolve in 15 kg of 94% acid and all of the sulfur dioxide leaves the solution, what is the strength of the acid in the which the copper sulfate is dissolved? (All acid compositions are in mass %). **(4 points)**